

SCENARIO CONTEXT: Investigate a Simulated Compromise from Phishing Email to Endpoint Hardening.

Purpose:

The project forces the student to apply every learning outcome listed — encryption, hashing & digital signatures, password hygiene, phishing analysis, DNS/VPN/Tor use, and systematic threat-hunting with monitoring tools. The student will produce a complete security-assessment report and a set of mitigation recommendations.

Background Details:

Your organization has just received an email that appears to be from the Finance department. The email contains a PDF attachment titled “Invoice_2024_Q2.pdf”. Suspicious behavior on a user’s workstation has been observed. A new service is running, unusual TCP connections appear, and a custom script hiding malicious code.

Your role as a Security Analyst:

1. Confirm that the email is phishing.
2. Verify the authenticity and integrity of the attached PDF. Find any flags if present.
3. Show that sensitive documents on the compromised workstation are encrypted.
4. Produce a structured threat-hunting report and a hardening plan.